

Exploring One Handed Signing During Driving for Interacting with In-vehicle Systems for Deaf and Hard of Hearing Drivers

Marie Lee
School of Information
Rochester Institute of Technology
Rochester, New York, USA
ml1225@rit.edu

Ziming Li
School of Information
Rochester Institute of Technology
Rochester, New York, USA
zl1398@rit.edu

Wendy Dannels
NTID Deaf Health Care and
Biomedical Science Hub
Rochester Institute of Technology
Rochester, New York, USA
w.dannels@rit.edu

Tae Oh
School of Information
Rochester Institute of Technology
Rochester, New York, USA
thoics@rit.edu

Roshan L Peiris
School of Information
Rochester Institute of Technology
Rochester, New York, USA
roshan.peiris@rit.edu

Abstract

Voice user interfaces (VUIs) offer safer interactions for drivers but remain largely inaccessible to Deaf or hard of hearing (DHH) individuals. Thus, our study explores the use of one-handed sign language and fingerspelling as alternative communication methods for DHH drivers when interacting with vehicle systems. Utilizing a Wizard of Oz (WoZ) approach, we investigated user preferences for three camera positions to detect sign language and feedback methods during interaction. Eighteen DHH participants engaged in a series of tasks within a driving simulator, including inputting text messages and modifying navigation destinations, with real-time interpretation provided by an expert signer through the WoZ setup. Our findings revealed a strong preference for the Center View (central vent area) and showed that the majority of participants favored a combination of one-handed signing and fingerspelling (15 of 18). Additionally, visual feedback was found to be effective and preferred for navigation during the WoZ driving simulation.

CCS Concepts

• **Human-centered computing** → **Empirical studies in accessibility**.

Keywords

Automotive user interface, Accessibility, American Sign Language, One-Handed Signing, Alternative Voice User Interface, Deaf and Hard of Hearing, Wizard of Oz, Driving

ACM Reference Format:

Marie Lee, Ziming Li, Wendy Dannels, Tae Oh, and Roshan L Peiris. 2025. Exploring One Handed Signing During Driving for Interacting with In-vehicle Systems for Deaf and Hard of Hearing Drivers. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*,

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719868>

April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages.
<https://doi.org/10.1145/3706599.3719868>

1 Introduction

Voice user interfaces (VUIs) in vehicles have gained significant popularity, with millions of users by early 2020 and a steadily growing user base since 2018 [20]. These systems, often integrated with digital assistants on smartphones and smart home devices such as Google Home and Amazon Echo, enabling interaction through spoken commands for tasks like composing text messages, setting navigation destinations, and controlling in-vehicle entertainment. This hands-free and eyes-free interaction is designed to support drivers in maintaining attention on the road while operating various vehicle functions [3, 27, 29].

Despite their advantages, VUIs present notable accessibility challenges for Deaf and hard of hearing (DHH) drivers due to their reliance on spoken input and auditory feedback [22]. Environmental factors, such as traffic noise and in-vehicle conversations, further complicate interaction and voice feedback recognition. Research highlights the barriers DHH users face when accessing VUIs [4, 34], emphasizing the need for alternative interaction modalities. Although accessible alternatives, such as sign language and augmented reality (AR) interfaces, have been explored [13, 26], these studies often focus on static or non-driving contexts. The unique challenges of implementing these alternatives in dynamic driving environments—particularly regarding safety, usability, and maintaining driver attention—remain underexplored.

To address this, in this study, we conducted a preliminary examination on the feasibility of using sign language, specifically one handed signing and fingerspelling, to interact with in-vehicle VUIs. Our motivation is rooted in prior studies showing that DHH users prefer American Sign Language (ASL) and Text-to-Speech (TTS) for interacting with voice-activated agents [30], while also highlighting that DHH signers often resort to one-handed signing when one hand is occupied [24] and use fingerspelling to convey or clarify specific words during conversations [15, 18]. Additionally, the fixed environment of a vehicle and already available in-vehicle cameras and gesture sensors create opportunities for gesture-sensing systems [8], which could leverage sign language as a natural and

accessible input modality. However, the implementation of such systems must address challenges related to safety and multitasking (e.g. drivers managing the steering wheel, maintaining focus on the road, and using sign language simultaneously face unique demands that do not arise in stationary contexts) and, consider additional factors such as camera placement for sign language input and feedback mechanisms.

Therefore, this study investigates the usability of one-handed signing and fingerspelling for tasks such as selecting destinations and entering messages through a Wizard-of-Oz (WoZ) study set up. By assessing their feasibility in dynamic driving conditions, it highlights accessibility and safety concerns in current voice-reliant in-vehicle interfaces and explores practical alternatives for DHH drivers.

In this work, we primarily focus on the following research questions:

- **RQ1a** What are DHH drivers' preferences when using one-handed signing and fingerspelling while driving?
- **RQ1b** Which camera angle provides the most comfortable and effective hand positioning for communication with in-vehicle systems?
- **RQ2:** What type of feedback do DHH drivers prefer when confirming the receipt of their commands or messages?

Our contributions are as follows:

- (1) This study explores one-handed sign language including finger spelling as an alternative to VUIs for in-vehicle interaction. Using a WoZ driving simulator with 18 DHH participants, it investigates user preferences and optimal camera placements for capturing input interactions.
- (2) Findings from the study are synthesized into a set of guidelines detailing DHH drivers' preferred methods of receiving feedback during in-vehicle interactions, focusing the effectiveness of visual and haptic modalities.

2 Related Work

2.1 Challenges in Speech Recognition and Accessibility

According to Alvarez et al. [2], the use of modern VUIs designed to effectively process natural language has enhanced the usability of speech recognition in vehicle systems. Despite these advancements, numerous studies have identified significant usability limitations, especially regarding accessibility and safety concerns. For example, misinterpretation or failure to recognize spoken commands can cause distractions in the driver, presenting considerable safety risks, as noted by Kim et al. [19]. Martelaro et al. [25] highlighted the impact of different levels of support from voice assistants and road context alerts on driver multitasking abilities, underscoring the need for reliable speech recognition to reduce cognitive overload. Compared to previous studies on smart devices, only a few have focused on the accessibility of VUIs in in-vehicle systems, particularly for people with disabilities. According to Blair et al. [4], DHH users face substantial challenges with speech recognition technology, including difficulties in processing voice pitch variations and an excessive dependence on visual feedback, such as text or graphical

notifications. In a similar context, Glasser's studies [11–13] demonstrate the obstacles encountered by DHH communities in utilizing automatic speech recognition systems, underscoring the need for alternative interaction methods. Also, Lee et al. [23] highlights the distinct challenges faced by DHH users when using ride sharing platforms, including communication issues with drivers and the dependence on visual notifications for app interactions, another study highlights visual communication requirements when using sign language [1]. These issues further emphasize the need for accessible and multi modal designs that reduce the reliance on vocal or exclusively visual feedback.

2.2 Advances and Challenges in Sign Language Recognition: Technical Barriers, Dataset Gaps, and the Impact of Viewing Angles

Recent advances in sign language recognition (SLR) have made significant strides in handling large vocabularies of both one- and two-handed signs, yet several challenges remain. One major hurdle is the capture of fine-grained finger movements and arm movements. As Wang et al. [36] observed, many systems like 'SignSpeaker' struggle to detect subtle finger movements using a single smartwatch accurately. Two-handed signs require synchronized tracking of both arms, often requiring additional sensors for accurate recognition. Moreover, continuous SLR struggles with segmenting sign sentences due to transitional movements and varying signal strengths. Beyond these technical issues, the field of sign language processing is dominated by broader challenges. According to Bragg et al. [5], the lack of high-quality public datasets and interdisciplinary expertise in Deaf studies, linguistics, Natural Language Processing (NLP), computer vision, and HCI hinders the development of scalable, accurate, and practical SLR systems. In this context, recent research by Shi et al. [32] focuses on detecting fingerspelling, a critical component of sign language where words are signed letter by letter. The study highlights the need to accurately identify time spans in signing videos containing fingerspelling for precise recognition and transcription. It also explores challenges in recognizing large sign language vocabularies, focusing on refined movements and articulated gestures essential for effective fingerspelling recognition.

2.3 Inclusive Interaction Design and Gesture-Based Interfaces in Automotive Systems

Designing an inclusive in-vehicle systems is essential to address the accessibility limitations of speech recognition technology, which is further hindered by environmental factors like traffic noise and in-vehicle conversations. Tobisch et al. [33] highlighted usability improvements in VUIs under uncertain input conditions, while gesture-based controls [8] and alternative input methods [16] offer potential solutions. Integrating SLR with multimodal inputs enhances both inclusivity and usability for DHH drivers. Recent research has increasingly focused on gesture-based user interfaces in this context. For instance, Graichen[14] highlighted the potential of gestures to improve user experiences, where participants preferred gesture-based interaction over touch-based interaction regardless

of task complexity. Fariman et al. [9] developed a user-defined gesture vocabulary to control vehicle climate systems, emphasizing adaptable interfaces. Gesture interaction has also been proposed as an alternative to voice recognition, with Cui et al. [7] suggesting surfaces near the driver for initiating gestures. Advances in mid-air gestures, such as those captured by sensors or enhanced through smart rings, shift control to the driver's fingers, offering safer and more intuitive in-vehicle interaction [10]. Shakeri et al. [31] further explored ultrasound haptic feedback to improve control via tactile sensations, enhancing accessibility and safety. Additionally, methods such as audio haptic interfaces have been adopted to present additional information through sounds for DHH users [35] that could enhance the space for feedback options.

3 Methods

To address RQ1a and RQ1b, we conducted a task-based study in which participants used one-handed ASL and fingerspelling to input commands into a WoZ system. The setup included a driving simulator, with a proficient signer hidden from the participants, who translated commands in real time via a video feed. The translated commands were displayed on a tablet integrated into the participant's setup (see Figure 1(a)). To evaluate the impact of camera placement, we tested three angles for capturing signing: Armrest View (low angle, Figure 1(b)), Center View (Figure 1(c)), and Ceiling View (high angle, Figure 1(d)). These camera angles and placement were determined based on existing camera and sensor placements in current vehicles for various purposes such as gesture sensing.

Participants were instructed to sign commands, such as inputting an address or text message, while operating the driving simulator. These commands simulated typical in-vehicle voice tasks and included a mix of random words and ASL signs to assess system usability. After signing, participants could confirm or re-enter commands using the tablet display. Their preferences and experiences were gathered through questionnaires and interviews.

For RQ2, participants provided feedback on their preferred methods and locations for receiving in-vehicle feedback, including visual and haptic options. While the visual feedback was shown on the tablet in the WoZ setup, the haptic feedback was presented as different vibration patterns presented through an audio haptic feedback module (Acouve Vp2) used in previous research [6, 35] that was initially attached to the steering wheel. Participants were able to freely explore these methods such as the modalities and the location of feedback during the evaluation. This data was collected through multiple-choice questions and interviews to better understand their sensory preferences and identify effective feedback methods.

3.1 Participants

The study included participants who met the following criteria: they were at least 18 years old, communicated using ASL, which was required to input commands, and had a valid driver's license due to the driving-focused nature of the research. A total of 18 participants (10 female and 8 male) were recruited through advertisements via flyers, emails, and word of mouth. All participants communicated in ASL, 14 identified as deaf and 4 as hard of hearing. The age range of the participants was 18 to 60 years (Mean = 23, Median = 21.5), including one left-handed individual.

3.2 Study Design

3.2.1 Equipment. The study employed a driving simulator setup designed to replicate a typical automobile interior (see Figure 1). A 43-inch TV displayed the CityCarDriving simulator software¹, while a 10.4-inch tablet simulated the vehicle's center console, displaying tasks and outputs in real time. Participants used a Logitech G29 steering wheel and pedals in a highway driving scenario. A mobile phone camera mounted on a flexible tripod captured one-handed sign language and fingerspelling inputs, allowing for adjustable angles as required. Pilot testing validated the setup's ability to capture detailed movements without interfering with the driving simulation. For the WoZ methodology, a proficient signer, positioned separately, translated participants' signing commands via a live Zoom feed² and entered them into the system. The tablet served as the driver's primary interface, displaying real-time feedback from the signing interpretations to ensure seamless communication during the simulation.

3.2.2 Post Questionnaire and Survey. A 5-point Likert scale (ranging from Strongly Agree to Strongly Disagree) was used across ten questions, with responses scaled and normalized to a 100-point score for comparative analysis. This normalization allowed for a standardized assessment of participant preferences and perceptions across different conditions.

3.3 Procedure

The 90-minute study was conducted in two sessions within a simulated driving environment to evaluate SL use across three camera angles (center, armrest, and ceiling), and a second session to evaluate the feedback methods (visual and haptics). Participants first reviewed consent forms and attended an orientation, where a proficient signer and moderator guided them through the procedures by presenting slides. They then practiced one-handed signing and fingerspelling using words displayed on tablet screens to familiarize themselves with the tasks while driving. In the first session, participants completed text messaging and navigation tasks while driving at 45 mph on the simulator. Input methods (fingerspelling or one-handed signing) and camera positions were randomly assigned on the tablet screen. A skilled signer translated participants' sign language in real time as they entered commands into the system, which were then displayed as text on the tablet monitor for driver verification. Participants completed a survey on input methods and camera angles before taking a break and switching camera positions. The second session focused on feedback methods, gathering participants' preferences for visual versus haptic feedback and their preferred feedback locations through structured questionnaires and semi-structured interviews, addressing RQ2.

4 Results

4.1 Quantitative Results

4.1.1 Preferences on camera angles. Figure 2 describes empirical findings on the perceived levels of comfort, safety, and efficiency as reported by participants in response to three different camera angles used during driving simulations. The survey found that 15

¹https://store.steampowered.com/app/493490/City_Car_Driving/

²<https://zoom.us/>

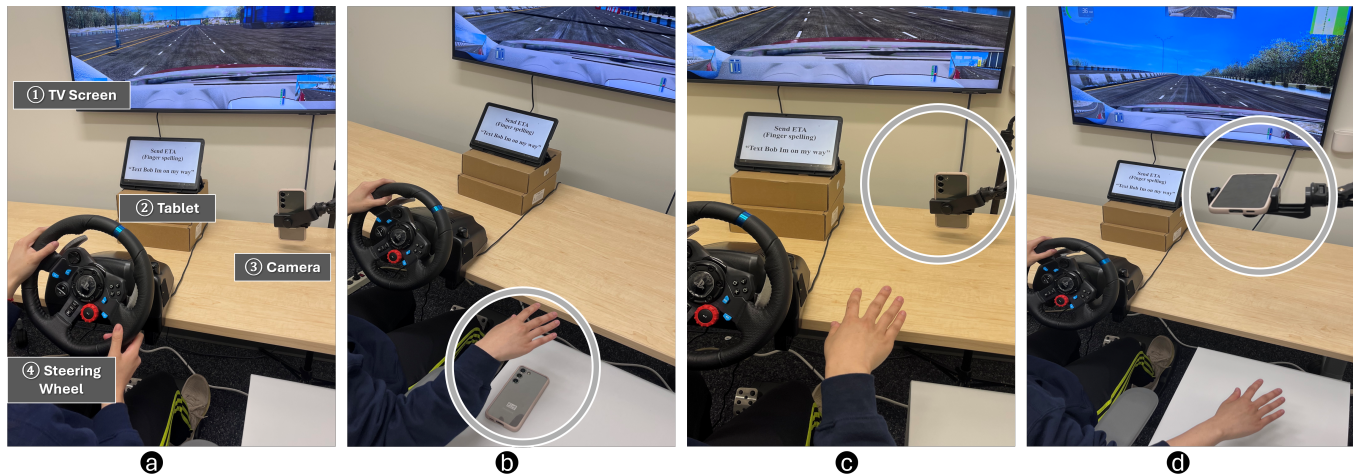


Figure 1: The study setup and camera positioning for the driving simulation: (a) Overview of the driving simulation setup includes: (1) A TV screen displaying the driving simulation environment; (2) A tablet simulating a vehicle's dashboard with task displays; (3) A mobile phone camera positioned to record hand gestures; (4) A steering wheel and pedals for driving controls. (b) Armrest view: A low-angle camera capturing hands above positioned by the armrest. (c) Center view: A camera at an intermediate height records gestures directly from the front central vent area. (d) Ceiling view: A high-angle camera positioned above captures hand gestures below.

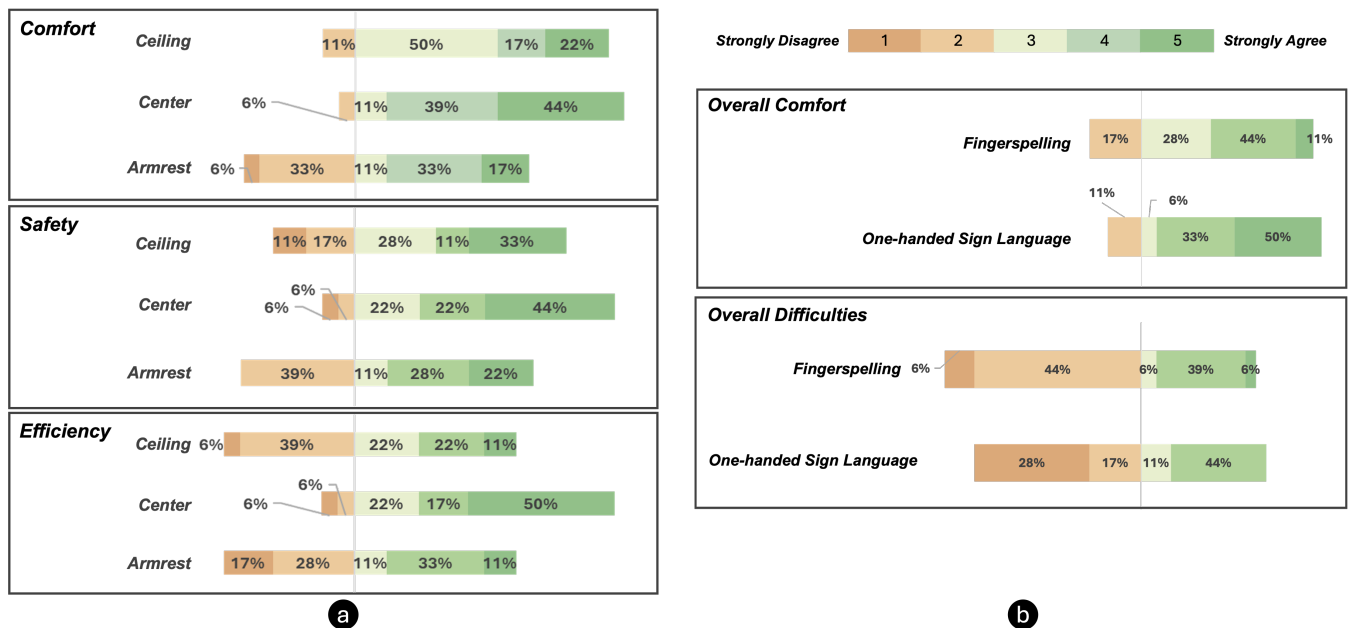


Figure 2: (a) Participant preference response rates for Comfort, Safety, and Efficiency. Preferences were assessed using a five-point Likert scale at each camera location. Each of the three rows in this figure represents a different camera angle, accompanied by bar graphs illustrating. (b) Responses for 'Overall Comfort' and 'Overall Difficulty' assessed using a five-point Likert scale, related to input methods of fingerspelling and one-handed sign language. Vertical lines in the bar graphs distinguish between the first two disagreement levels and those beyond the neutral midpoint.

out of 18 participants felt comfortable using the center view camera, with mean ratings for the Ceiling, Center, and Armrest views were 3.5, 4.22, and 3.22 (with 5 being 'Very Comfortable'). The center view was rated as the most secure, with 12 participants labeling it 'Very Safe' or 'Safe'. The views from the ceiling and armrest were considered secure by 8 and 9 respondents, respectively. The central camera position was identified as the most efficient for task performance, followed by the armrest view by 8 participants. The survey results indicated a strong preference for the center view position, with 15 out of 18 participants ranking it as their top choice. The ceiling view position was favored by 2 respondents, and only 1 participant considered the armrest position as the best choice, making it the least preferred position.

4.1.2 Accessibility and usability of the proposed methods. When we asked about their overall opinion on using one-handed signing or fingerspelling to interact with in-vehicle systems, 16 participants out of 18 answered they were generally satisfied. P1 highlighted its natural integration while driving, *"I really liked that I didn't have to move my hand too far away from the wheel, making it feel like a natural conversation."* P12 stated, *"I'd prefer to use a mix of fingerspelling and signs. Avoid need to fingerspell filler words and allow signs of common words when fingerspelling specific addresses."* This highlights the preference for minimizing unnecessary fingerspelling, a concern echoed by P18, who noted the increased cognitive load of fingerspelling long commands while driving. Despite limitations of the armrest camera angle, some found it useful for fingerspelling-based input. P4 observed, *"If I were just to fingerspell then this position (armrest view) is good for that but it isn't good if i were to sign as i cant use my body for it."*

4.1.3 Concerns and suggestions. Common concerns and suggestions have emerged from participants about diverted attention when adjusting signing for camera angles, as well as poor visibility for certain gestures, like the 'My' sign. The P4's opinion supports that *"When I went to sign 'my' with the ceiling view angle and armrest angle, it was out of the camera, and I had to adjust to fingerspelling it, and that felt unnatural. This forced signers to adopt less natural stances, disrupting the natural flow of conversation."* Fingerspelling posed challenges for complex inputs, requiring visual focus that could impact driving safety. Adjusting for camera placement further disrupted the natural signing flow, reinforcing the importance of a centrally positioned camera.

4.2 Findings on Feedback Methods

4.2.1 Preference for sensory modalities. There are various factors to consider in maintaining stability and communication efficiency while driving. Previous studies have demonstrated that diverting one's gaze from the road to interact with vehicle controls can significantly compromise driving stability [17]. To maintain driving stability, participants preferred visual feedback positioned near the line of sight. P7 suggested, *"Dashboard, near to the windscreen. So that the eye level can see the phone and the road same time. That way we can concentrate on our driving while interacting safely."* P3 also highlighted the potential of haptic feedback for volume control and sign-based cruise control. P15 emphasized customization: *"...having a screen close to the road inside the vehicle would be beneficial. Haptic*

feedback could be effective for volume control, ... using cruise control technology and signing to a camera to make driving easier."

4.2.2 Preferences on combinations. While most participants favored visual feedback, four preferred visual and haptic feedback. P10 recommended a center view for visual and an armrest angle for haptic. P6 proposed captioning glasses similar to the ones used in movie theater.

4.2.3 Factors influencing preferences. Various factors influence preferences for feedback location and modality, as identified through analysis of open-ended responses. One such aspect is the ability to view feedback easily without causing distractions while driving. P1 observed, *"Having close proximity to the feedback makes me feel safer since I don't have to take my eyes off the road for long periods."* Another factor is the ease of accessing controls and familiarity with areas such as the center vent or dashboard. For example, P10 stated, *"Being able to sign and fingerspell with ease while minimizing the impact on my hand, wrist, and shoulder is important."*

5 Discussion

This study examined the usability of one-handed ASL and fingerspelling as alternatives to traditional in-vehicle VUIs using a driving simulation with the WoZ method, involving 18 DHH participants. User preferences and system usability were assessed across three camera angles (ceiling, center, and armrest) while performing real-world input tasks. Findings showed that the center camera, capturing the face and upper body, was the most preferred and efficient for RQ1. Participants also favored a combination of one-handed signing and fingerspelling over fingerspelling alone, as it was less physically and cognitively demanding while driving. For RQ2, survey results indicated a strong preference for visual feedback in command confirmation. Most participants found it is particularly beneficial for navigation and safety alerts to prevent road accidents. Overall, this study explored the feasibility of in-vehicle one-handed interactions for DHH drivers, an area less examined compared to traditional fingerspelling and two-handed signing.

5.1 Safety Implications of In-Vehicle Technologies

The findings of our study emphasize the critical role of camera placement: participants expressed concerns about the effectiveness of one-handed sign language when the camera angle is either too high or too low. This poses a significant safety risk, as drivers must be able to see their hand gestures within the camera's field of view without looking away from the road. Any distraction from road observation can seriously compromise driver safety. To mitigate these issues and enhance communication safety, future designs should include robust feedback mechanisms that adapt to the dynamics of ASL motion. Recent research supports the use of haptic feedback to reduce the need for visual confirmation when drivers interact with touchscreen interfaces [28]. Furthermore, studies have shown that integrating voice and tactile feedback not only accelerates task completion times compared to voice-only feedback but also maintains driving performance and does not increase cognitive load [17]. Based on these findings, we suggest for the integration of tactile feedback, enabling DHH drivers to receive prompt and effective

feedback with less difficulty. This approach aims to enhance both the safety and functionality of their interactions.

5.2 Accessibility of Using Sign Language While Driving

Our study did not examine the technical implementation of detecting sign language or fingerspelling during driving. However, prior research demonstrates the feasibility of recognizing fingerspelling, as seen in technologies like the ASL one-handed spelling game³. While effective, broader recognition of one-handed sign language remains challenging due to the complexity of subtle hand movements, shapes, positions, and overlapping finger motions. Additionally, sign language-based interactions introduce linguistic structure and semantic meaning, which differ from general gesture recognition systems that prioritize simplicity over syntax. Automatic sign language recognition (SLR) is particularly intricate, requiring computer vision algorithms to accommodate individual differences in signing styles [21]. Future research should address these challenges by developing diverse datasets that capture a wide range of signers and optimizing for key variables such as lighting conditions and camera angles, which are especially problematic in dynamic driving environments, as highlighted by Bragg et al. [5]. Ensuring robustness across these factors is crucial for achieving reliable sign language recognition in vehicular settings. Therefore, a holistic approach is essential to advance robust and inclusive sign language recognition technologies.

5.3 Limitations and Future Work

Our study identifies several limitations that should be addressed in future research. First, the sample size was limited, affecting the validity and generalizability of the results. Future studies should include participants with diverse age groups, backgrounds, and sign language proficiency to enhance representativeness.

Second, as the study was conducted in a simulated environment using the WoZ method, it did not fully replicate real-world driving conditions such as driving in different environments (in the night, while sunny or under shade) and under different conditions (when raining, heavy traffic, etc.). Future research should prioritize immersive, realistic scenarios that incorporate variations in speed, traffic signals, and potential distractions to better assess system performance. Testing under realistic conditions may also open discussions on other safety related matters such as requiring to disable the system under certain driving conditions for safety or potential driver distraction if users adjust their signing to fit within the camera frame.

Third, technical limitations such as fixed camera positions, inconsistent lighting, and occasional streaming interruptions highlight the need for real-time adaptive data processing. Future systems should dynamically adjust to the movement of DHH users to enhance precision and functionality.

Finally, while participant training sessions helped mitigate driving challenges, the study did not directly assess driver distractions. Future research should focus on evaluating distraction levels and safety implications to better understand the system's impact on driver concentration and overall stability.

³<https://deafchildren.org/knowledge-center/asl-resources/asl-fingerspelling-app/>

6 Conclusion

This study explored the feasibility of using one-handed sign language and fingerspelling for in-vehicle interactions among DHH participants as an alternative to voice recognition-based vehicle control, along with their preferred feedback methods. Eighteen DHH drivers participated in a task-based experimental study using a WoZ driving simulation. Participants controlled an in-vehicle system from three camera positions ceiling, center, and armrest performing tasks with one-handed signing and fingerspelling. The central camera position, capturing the face and upper body, was preferred by most participants. Additionally, participants favored combining one-handed signing with fingerspelling due to lower physical and cognitive demands while driving, and visual feedback was effective for navigation and safety alerts. These findings highlight the feasibility and benefits of one-handed in-vehicle interactions for DHH drivers and underscore the importance of developing accessible automotive interfaces tailored to their needs.

References

- [1] Chanchal Agrawal and Roshan L Peiris. 2021. I See What You're Saying: A Literature Review of Eye Tracking Research in Communication of Deaf or Hard of Hearing Users. In *Proceedings of the 23rd International ACM SIGACCESS Conference on Computers and Accessibility* (Virtual Event, USA) (ASSETS '21). Association for Computing Machinery, New York, NY, USA, Article 41, 13 pages. <https://doi.org/10.1145/3441852.3471209>
- [2] Ignacio Alvarez, Aqueasha Martin, Jerone Dunbar, Joachim Taiber, Dale-Marie Wilson, and Juan E Gilbert. 2010. Voice interfaced vehicle user help. In *Proceedings of the 2nd International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. 42–49.
- [3] Adriana Barón and Paul Green. 2006. Safety and usability of speech interfaces for in-vehicle tasks while driving: A brief literature review. (2006).
- [4] Johnna Blair and Saeed Abdullah. 2020. It Didn't Sound Good with My Cochlear Implants: Understanding the Challenges of Using Smart Assistants for Deaf and Hard of Hearing Users. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 4 (2020), 1–27.
- [5] Danielle Bragg, Oscar Koller, Mary Bellard, Larwan Berke, Patrick Boudreaux, Annelies Braffort, Naomi Caselli, Matt Huenerfauth, Hernisa Kacorri, Tessa Verhoef, et al. 2019. Sign language recognition, generation, and translation: An interdisciplinary perspective. In *Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility*. 16–31.
- [6] Pratheep Kumar Chelladurai, Ziming Li, Maximilian Weber, Tae Oh, and Roshan L Peiris. 2024. SoundHapticVR: Head-Based Spatial Haptic Feedback for Accessible Sounds in Virtual Reality for Deaf and Hard of Hearing Users. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility* (St. John's, NL, Canada) (ASSETS '24). Association for Computing Machinery, New York, NY, USA, Article 31, 17 pages. <https://doi.org/10.1145/3663548.3675639>
- [7] Zhitong Cui, Hebo Gong, Yanan Wang, Chengyi Shen, Wenying Zou, and Shijian Luo. 2021. Enhancing interactions for in-car voice user interface with gestural input on the steering wheel. In *13th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. 59–68.
- [8] Tanja Döring, Dagmar Kern, Paul Marshall, Max Pfeiffer, Johannes Schöning, Volker Gruhn, and Albrecht Schmidt. 2011. Gestural interaction on the steering wheel: reducing the visual demand. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, BC, Canada) (CHI '11). Association for Computing Machinery, New York, NY, USA, 483–492. <https://doi.org/10.1145/1978942.1979010>
- [9] Hessam Jahani Fariman, Hasan J Alyamani, Manolya Kavakli, and Len Hamey. 2016. Designing a user-defined gesture vocabulary for an in-vehicle climate control system. In *Proceedings of the 28th Australian Conference on Computer-Human Interaction*. 391–395.
- [10] Bogdan-Florin Gheran and Radu-Daniel Vatavu. 2020. From controls on the steering wheel to controls on the finger: using smart rings for in-vehicle interactions. In *Companion Publication of the 2020 ACM Designing Interactive Systems Conference*. 299–304.
- [11] Abraham Glasser. 2019. Automatic speech recognition services: Deaf and hard-of-hearing usability. In *Extended abstracts of the 2019 CHI conference on human factors in computing systems*. 1–6.
- [12] Abraham Glasser, Kesavan Kushalnagar, and Raja Kushalnagar. 2017. Deaf, hard of hearing, and hearing perspectives on using automatic speech recognition in conversation. In *Proceedings of the 19th International ACM SIGACCESS Conference*

- on *Computers and Accessibility*. 427–432.
- [13] Abraham Glasser, Matthew Watkins, Kira Hart, Sooyeon Lee, and Matt Huenerfauth. 2022. Analyzing Deaf and Hard-of-Hearing Users' Behavior, Usage, and Interaction with a Personal Assistant Device that Understands Sign-Language Input. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–12.
 - [14] Lisa Graichen and Matthias Graichen. 2024. Do users desire gestures for in-vehicle interaction? Towards the subjective assessment of gestures in a high-fidelity driving simulator. *Computers in Human Behavior* (2024), 108189.
 - [15] Saad Hassan, Abraham Glasser, Max Shengelia, Thad Starner, Sean Forbes, Nathan Qualls, and Sam S. Sepah. 2023. Tap to Sign: Towards using American Sign Language for Text Entry on Smartphones. *Proc. ACM Hum.-Comput. Interact.* 7, MHCI, Article 227 (sep 2023), 23 pages. <https://doi.org/10.1145/3604274>
 - [16] Hansjörg Hofmann, Vanessa Tobisch, Ute Ehrlich, André Berton, and Angela Mahr. 2014. Comparison of speech-based in-car HMI concepts in a driving simulation study. In *Proceedings of the 19th international conference on intelligent user interfaces*. 215–224.
 - [17] Jungun Jung, Sangyoon Lee, Jiwoo Hong, Eunhye Youn, and Geehyuk Lee. 2020. Voice+ tactile: Augmenting in-vehicle voice user interface with tactile touchpad interaction. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–12.
 - [18] Jonathan Keane, Diane Brentari, and Jason Riggie. 2013. Coarticulation in ASL fingerspelling. In *Proceedings of the North East Linguistic Society*. North East Linguistic Society.
 - [19] Jihyun Kim, Meuel Jeong, and Seul Chan Lee. 2019. "Why did this voice agent not understand me?" error recovery strategy for in-vehicle voice user interface. In *Proceedings of the 11th International Conference on Automotive User Interfaces and Interactive Vehicular Applications: Adjunct Proceedings*. 146–150.
 - [20] Bret Kinsella and Ava Mutchler. 2020. In-car voice assistant consumer adoption report.
 - [21] Dimitrios Konstantinidis, Kosmas Dimitropoulos, and Petros Daras. 2018. SIGN LANGUAGE RECOGNITION BASED ON HAND AND BODY SKELETAL DATA. In *2018 - 3DTV-Conference: The True Vision - Capture, Transmission and Display of 3D Video (3DTV-CON)*. 1–4. <https://doi.org/10.1109/3DTV.2018.8478467>
 - [22] Sanskriti Kumar, Wendy Dannels, Tae Oh, and Roshan L. Peiris. 2025. Exploring Accessibility of Voice Controlled Interfaces and Gesture Interactions in Vehicles for Secondary Driving Tasks for Deaf and Hard of Hearing Drivers (CHI EA '25). Association for Computing Machinery, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3706599.3719843>
 - [23] Sooyeon Lee, Björn Hubert-Wallander, Molly Stevens, and John M Carroll. 2019. Understanding and designing for deaf or hard of hearing drivers on Uber. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–12.
 - [24] Kelly Mack, Danielle Bragg, Meredith Ringel Morris, Maarten W. Bos, Isabelle Albi, and Andrés Monroy-Hernández. 2020. Social App Accessibility for Deaf Signers. *Proc. ACM Hum.-Comput. Interact.* 4, CSCW2, Article 125 (oct 2020), 31 pages. <https://doi.org/10.1145/3415196>
 - [25] Nikolas Martelaro, Jaime Teevan, and Shamsi T Iqbal. 2019. An exploration of speech-based productivity support in the car. In *Proceedings of the 2019 CHI conference on human factors in computing systems*. 1–12.
 - [26] Roshan Mathew, Garreth W. Tigwell, and Roshan L. Peiris. 2024. Deaf and Hard of Hearing People's Perspectives on Augmented Reality Interfaces for Improving the Accessibility of Smart Speakers. In *Universal Access in Human-Computer Interaction*, Margherita Antona and Constantine Stephanidis (Eds.). Springer Nature Switzerland, Cham, 334–357.
 - [27] Matthias Peissner, Vanessa Doebler, and Florian Metze. 2011. Can voice interaction help reducing the level of distraction and prevent accidents. *Meta-Study Driver Distraction Voice Interaction* 24, 5 (2011).
 - [28] Matthew J Pitts, Gary Burnett, Lee Skrypchuk, Tom Wellings, Alex Attridge, and Mark A Williams. 2012. Visual-haptic feedback interaction in automotive touchscreens. *Displays* 33, 1 (2012), 7–16.
 - [29] Martin Porcheron, Joel E Fischer, Stuart Reeves, and Sarah Sharples. 2018. Voice interfaces in everyday life. In *proceedings of the 2018 CHI conference on human factors in computing systems*. 1–12.
 - [30] Jason Rodolitz, Evan Gambill, Brittany Willis, Christian Vogler, and Raja Kushalnagar. 2019. Accessibility of voice-activated agents for people who are deaf or hard of hearing. *Journal on Technology and Persons with Disabilities* 7 (2019), 144–156.
 - [31] Gözel Shakeri, John H Williamson, and Stephen Brewster. 2018. May the force be with you: Ultrasound haptic feedback for mid-air gesture interaction in cars. In *Proceedings of the 10th international conference on automotive user interfaces and interactive vehicular applications*. 1–10.
 - [32] Bowen Shi, Diane Brentari, Greg Shakhnarovich, and Karen Livescu. 2021. Fingerspelling detection in american sign language. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 4166–4175.
 - [33] Vanessa Tobisch, Markus Funk, and Adam Emfield. 2020. Dealing with input uncertainty in automotive voice assistants. In *12th international conference on automotive user interfaces and interactive vehicular applications*. 161–168.
 - [34] Alessandro Diogo Vieira, Higor Leite, and Ana Vitória Lachowski Volochchuk. 2022. The impact of voice assistant home devices on people with disabilities: A longitudinal study. *Technological Forecasting and Social Change* 184 (2022), 121961.
 - [35] Yiwen Wang, Ziming Li, Pratheep Kumar Chelladurai, Wendy Dannels, Tae Oh, and Roshan L Peiris. 2023. Haptic-Captioning: Using Audio-Haptic Interfaces to Enhance Speaker Indication in Real-Time Captions for Deaf and Hard-of-Hearing Viewers. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 781, 14 pages. <https://doi.org/10.1145/3544548.3581076>
 - [36] Zhibo Wang, Tengda Zhao, Jinxin Ma, Hongkai Chen, Kaixin Liu, Huajie Shao, Qian Wang, and Ju Ren. 2020. Hear sign language: A real-time end-to-end sign language recognition system. *IEEE Transactions on Mobile Computing* 21, 7 (2020), 2398–2410.